



Original Article

Mathematical analysis of fractional order alcoholism model

Muhammad Sher^a, Kamal Shah^{a,b}, Muhammad Sarwar^a, Manar A. Alqudah^c,
Thabet Abdeljawad^{b,d,e,f,*}

^a Department of Mathematics, University of Malakand, Chakdara, Khyber Pakhtunkhwa, Pakistan

^b Department of Mathematics and Sciences, Prince Sultan University, P.O. Box 66833, 11586 Riyadh, Saudi Arabia

^c College of Science, Princess Nourah bint Abdulrahman University P. O. Box 84428, Riyadh 11671, Saudi Arabia

^d Department of Medical Research, China Medical University, Taichung 40402, Taiwan

^e Department of Mathematics Kyung Hee University, 26 Kyungheedaero-ro, Dongdaemun-gu, Seoul 02447, Republic of Korea

^f Department of Mathematics and Applied Mathematics, School of Science and Technology, Sefako Makgatho Health Sciences University, Ga-Rankuwa, South Africa

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ABSTRACT

In this manuscript, we are going to study a novel model of the dynamics of alcohol consumption under induced complications. The mentioned model is considered under the concept of conformable fractional order derivative (CFOD). Currently, most of real-world problems are considered under fractional order derivatives because of their stable and global behavior. First, we will investigate the model for qualitative theory including existence and uniqueness of solution and Ulam-Hyers stability. For qualitative theory, we will use fixed point theory. In addition, we use a numerical method to find the approximate solution of the proposed model. In the final part of the paper, we give a detailed discussion of its numerical results and its graphical presentation.

1. Introduction

It is well-known fact that misuse of alcohol increasing day by day which is very dangerous for the whole globe. It can be seen that the misuse of alcohol in the young generation is rapidly increasing because they used it for fun and enjoyment. In 2019, it was falsely recommended that alcohol can be used for the treatment of (Covid-19) disease. That is why its consumption during Covid-19 increased [1]. According to different sources alcohol abuse kills 2.5 million population per year [2]. It is estimated by WHO that, alcohol-abuse cause various types of disease like liver cancer, heart disease, brain disorder, high blood pressure, digestive problems, etc. In addition, alcohol dependence causes a person to experience withdrawal symptoms involving headache, nausea, and exhaustion when they are unable to consume alcohol. As alcohol consumption increases, the person can lose control as they get obsessed with drinking. He or she might experience blackouts, which are instances in which someone entirely forgets what happened while they were intoxicated even if they were awake at the time. Lastly, personality alterations take place. Alcoholism can make a person more violent and substantially impair their capacity to function (keep down a job or uphold friendships and family ties). Heavy drinkers may experience

seizures, hallucinations, panic attacks, confusion, and tremors (see [32]). Recently, in many places due to the misuse of drugs, various incidents occurs daily in which many people lose their lives. In addition, excessive misuse can seriously damage the stomach, liver, heart, brain, and nervous system which cause many other savior diseases in human body.

Since misuse of alcohol is a real- world problem. To understand the mechanism and procedure, how the drugs abuse occur and destroy the human life, researchers use various tools and analysis. Here, we remark that the area devoted to mathematical models of real-world problems plays a very crucial role in investigation and analyzing. For instance mathematical models play significant roles in controlling different diseases. In the past researchers have successfully attempted to model infectious diseases to predict the effects of epidemics in humans population but the world still struggles to handle such types of problems due to natural or artificial phenomena. To control such risk around the globe it is necessary for a mathematician to play their role (see [3–5]). Previously researchers have attempted to model different diseases including alcohol drinking. Here, we refer some well reputed work like [6,7]. Therefore, many researchers have used different types of mathematical models of infectious diseases to understand the transmission

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* Corresponding author at: Department of Mathematics and Sciences, Prince Sultan University, P.O. Box 66833, 11586 Riyadh, Saudi Arabia.

E-mail addresses: maalqudah@pnu.edu.sa (M.A. Alqudah), tabdeljawad@psu.edu.sa (T. Abdeljawad).

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mechanisms, and predictions and choose the best control strategies, for detail see [8,9] and reference therein. Here, we remark that authors [10] investigated the heroin addiction model. In the same way, bifurcation analysis, global stability, and numerical investigations have been performed for heroin epidemic models in [11–14]. In addition, authors [15] have studied global stability analysis for alcohol abuse model. Authors [16,17], studied a deterministic model for the dynamics of alcohol abuse in detail. They have formulated the model with five different compartments. The mentioned model is given by:

$$\begin{cases} P'(\eta) = b + \gamma_2 R - (\lambda + \mu)P \\ M'(\eta) = \lambda P - (\mu + \theta + \alpha_3)M, \\ H'(\eta) = \theta M - (\alpha_1 + \alpha_2 + \mu + \delta_1)H, \\ C'(\eta) = \alpha_1 H - (\gamma_1 + \mu + \delta_2)C, \\ R'(\eta) = \alpha_2 H + \alpha_3 M + \gamma_1 C - (\mu + \gamma_2)R. \end{cases} \quad (1)$$

Here, P, M, H, C , and R respectively represent potential drinkers, moderate alcohol abusers, heavy alcohol abusers, heavy alcohol abusers with induced complications, and the recovered class. the detail of various parameters used in model (1) is given as follow

It is important to note that previously and also currently the researchers model different biological and physical problems using integer order derivatives. But in the last two decades, fractional calculus has gotten more attention because of their application in the scientific world. Lots of work has been done by mathematicians to model different biological problems under fractional order derivatives and studied their various aspects. Using fractional order derivatives have numerous benefits. For instance the Caputo, and the Riemann–Liouville definitions have some nonlocal characteristics including preservation of past memory and future behavior which are inherited by these two fractional derivatives. Here it should be kept in mind that the said area has increasingly applied in engineering and science, including optics, signals processing, viscoelasticity, fluid mechanics, electrochemistry, biological population models, and electromagnetic. In order to better understand physical and engineering processes, differential operators with fractional order have been utilized to simulate them. Additionally, non-integer order systems have been used to investigate accurately the damping behaviors as discussed in [18]. In the same way, authors [19–23] recently used fractional-order derivative to investigate different neural network problems using bifurcation analysis. In additions authors [24,25] have studied fractional-order stage-structured predator–prey models using bifurcation and numerical analysis. In the same, authors [26] investigated chemical reaction model using fractional order delay differential equations. Recently, many researchers have applied the concept of fractional calculus in biological models as well. For instance, author [27] studied a biological model numerically using fractional order derivatives. Moreover, authors [28] used fractional order derivatives to study terrorism and corruption codynamics. Here, we refer some more useful investigations on the area of mathematical models using fractional calculus such as [29–31]. Due to wide applications and stable behavior of fractional order operators, in the last few decades, the said area has been considered very well for solutions to complex phenomena in different scientific disciplines. It has been observed that the calculus of fractional order gives more realistic results than integer order. Also fractional derivative is global in nature, which makes them more interesting tool than the classical one. For further analysis on fractional calculus, one may read [32].

Here it is interesting that fractional order derivative has no unique definition. Different mathematicians have defined it in different ways according to the need for real-world problems. In literature, we have various definitions like Reimann-Liouville, Caputo, Atangan-Baleanu, Caputo-Febrizio, Hilfer, conformable, etc. The most well-known definitions are Reimann-Liouville and Caputo fractional derivatives. To the best of our knowledge, these have been considered very well in the recent past. But these definitions do not assume the chain rule like integer order derivative. Here, we note that, in terms of physical

significance, the conformable fractional derivative can be a superior substitute for the traditional Riemann–Liouville and Caputo fractional derivative. Additionally, while conformable complies with the well-known product rule for two functions, all fractional derivatives do not. The considered operator, in our opinion, offers two benefits over the conventional fractional derivatives. First, the CFOD definition is straightforward and meets the majority of the requirements for the classical integral derivative, including linearity, the Rolle's theorem, the mean value theorem, the product rule, the quotient rule, the power rule, the chain rule, vanishing derivatives for constant functions, and so on. Second, the said derivative is very practical when used to model a variety of physical issues because it makes it simpler to numerically solve differential equations with CFOD than with Riemann–Liouville or Caputo fractional derivatives. For reader, we refer [33–35]. Also in CFOD, there are no memory terms and sometimes we take it as a local differential operator fractional order. In the recent past some work has been done by different researchers using this operator, for detail see [36,37]. Additionally, mathematical models with fractional order derivatives are naturally related to systems with memory involving in many biological systems.

From the above discussion, it is important to study various real-world problems under fractional differential operators is more realistic than using the traditional derivatives. So in this regard in the current manuscript, we generalize the acholic-abuse model (1) from integer order to fractional order. We consider the model under CFOD. Our considerer model is described as

$$\begin{cases} \mathcal{I}_{\eta_0}^r P(\eta) = b + \gamma_2 R - (\lambda + \mu)P, \\ \mathcal{I}_{\eta_0}^r M(\eta) = \lambda P - (\mu + \theta + \alpha_3)M, \\ \mathcal{I}_{\eta_0}^r H(\eta) = \theta M - (\alpha_1 + \alpha_2 + \mu + \delta_1)H, \\ \mathcal{I}_{\eta_0}^r C(\eta) = \alpha_1 H - (\gamma_1 + \mu + \delta_2)C, \\ \mathcal{I}_{\eta_0}^r R(\eta) = \alpha_2 H + \alpha_3 M + \gamma_1 C - (\mu + \gamma_2)R. \end{cases} \quad (2)$$

In addition, the initial conditions are described as

$$\begin{cases} P(0) = P_0 \geq 0, \\ M(0) = M_0 \geq 0, \\ H(0) = H_0 \geq 0, \\ C(0) = C_0 \geq 0, \\ R(0) = R_0 \geq 0. \end{cases} \quad (3)$$

As earlier discussed that fractional differential problems have greater degree of freedom. But on other hand it is very tedious job to obtain the exact analytical solution for every problem. Also, in many situation analytical and closed solutions of the aforementioned problems cannot generally be computed easily. Therefore, best numerical approximate tools are always needed. Researchers have developed various numerical tools to investigate the aforesaid area for simulations. Because, we can understand the solution behavior of such problems and exploring their applications Some useful numerical techniques been studied earlier for fractional order problems are refereed as [38–40]. On the other hand, the qualitative theory is a powerful tool to use to identify weather the consider problem with fractional order has a solution or not. Also, the existence of solution and its uniqueness is very important procedure. To obtain the required results, we usually use some fixed point theory and tools of functional analysis (see [41–45]). Additionally, stability is important consequence from optimization and numerical point of view to be studied for numerical solution of aforesaid problems. In this regards, Ulam-Hyers (UH) stability analysis plays significant roles to investigate the required aspects. Researchers in this regards have done valuable work for traditional problems of fractional calculus. For instance, we refer here some results as [46–48]. The existence theory and UH stability have been extended to problems involving CFOD (see [49,50]). First, we develop the qualitative theory using fixed point theory. In addition, we discuss UH stability of the model under our

consideration. To simulate our results, we also approximate the solution of the considered model (4) using numerical scheme given [51]. For existence theory and UH stability, we will use some tools from functional analysis. For an approximate solution, we will use Euler's method which is the most simple method in the literature. Also, the positivity and boundedness of solution has been proved following the procedure given in [52,53].

2. Fundamental results

In this part of our manuscript we recall some fundamental results. Some needful results are recalled from [33–37].

Definition 2.1. The CFOD of a function $\mathcal{W} : [\eta_0, \infty) \rightarrow \mathbb{R}$ with order $r \in (0, 1]$ is given by

$$\mathcal{I}_{\eta_0}^r \mathcal{W}(\eta) = \lim_{h \rightarrow \eta_0} \frac{\mathcal{W}(\eta + h\eta^{1-r}) - \mathcal{W}(\eta)}{h}, \text{ for every } \eta > \eta_0,$$

provided that if $\mathcal{I}_{\eta_0}^r \mathcal{X}(\eta) = \lim_{\eta \rightarrow \eta_0} D_r^{\eta_0} \mathcal{W}(\eta)$.

Definition 2.2. The conformable fractional integral of a function $\mathcal{W} : [\eta_0, \infty) \rightarrow \mathbb{R}$ with order $r \in (0, 1]$ is given by

$$\mathcal{I}_{\eta_0}^r \mathcal{W}(\eta) = \int_{\eta_0}^{\eta} (\psi - \eta_0)^{r-1} \mathcal{W}(\psi) d\psi, \text{ for every } \eta > \eta_0,$$

provided that the integral on the right exists.

Lemma 2.3. Under the continuity of $\mathcal{W} : [\eta_0, \infty) \rightarrow \mathbb{R}$, we have the given results.

$$\mathcal{I}_{\eta_0}^r [\mathcal{I}_{\eta_0}^r \mathcal{W}(\eta)] = \mathcal{W}(\eta), \quad r \in (0, 1], \text{ for every } \eta > \eta_0.$$

Lemma 2.4. In addition, we have the following result

$$\mathcal{I}_{\eta_0}^r [\mathcal{I}_{\eta_0}^r \mathcal{W}(\eta)] = \mathcal{W}(\eta) - \mathcal{W}(\eta_0), \quad r \in (0, 1].$$

3. Theoretical Analysis of the Proposed Model

3.1. Positivity and Boundedness

We follow the procedure described in [52,53], the positivity of the solution of the considered model can be deduced as

$$\begin{cases} \mathcal{I}_{\eta_0}^r P(\eta) \Big|_{P=0} = b + \gamma_2 R > 0, \\ \mathcal{I}_{\eta_0}^r M(\eta) \Big|_{M=0} = \lambda P \geq 0, \\ \mathcal{I}_{\eta_0}^r H(\eta) \Big|_{H=0} = \theta M \geq 0, \\ \mathcal{I}_{\eta_0}^r C(\eta) \Big|_{C=0} = \alpha_1 H \geq 0, \\ \mathcal{I}_{\eta_0}^r R(\eta) \Big|_{R=0} = \alpha_2 H + \alpha_3 M + \gamma_1 C \geq 0. \end{cases} \quad (4)$$

Hence, the proposed model is positive if the initial conditions are positive. In addition on every hyper plane bounding the non-negative, and the vector field points into $\mathbb{R}_+^5$. Further, the boundedness is given in next theorem.

Theorem 3.1. The consider model is bounded and the existence holds in the feasible region given by

$$\Omega = \left\{ (P, M, H, C, R) \in \mathbb{R}_+^5 : N = P + M + H + C + R \leq \frac{b}{\mu} \right\}. \quad (5)$$

Proof. Let N be the total population at time say η . Then, we have

$$N(\eta) = P(\eta) + M(\eta) + H(\eta) + C(\eta) + R(\eta). \quad (6)$$

Taking conformable derivative of fractional order $r > 0$ of (6), one has

$$\begin{aligned} \mathcal{I}_{\eta_0}^r [N(\eta)] &= \mathcal{I}_{\eta_0}^r [P(\eta)] + \mathcal{I}_{\eta_0}^r [M(\eta)] + \mathcal{I}_{\eta_0}^r [H(\eta)] + \mathcal{I}_{\eta_0}^r [C(\eta)] + \mathcal{I}_{\eta_0}^r [R(\eta)] \\ &= b - \mu P - \mu M - \mu H - \mu C - \mu R - \delta_1 H - \delta_2 C \\ &\leq b - \mu N(\eta). \end{aligned} \quad (7)$$

Taking conformable Laplace transform of both sides of (7), and using $\eta \rightarrow \infty$, we have

$$N(\eta) \leq \frac{b}{\mu}.$$

Hence the solution is bounded and we can express the feasible region as described in (5).

3.2. Equilibrium Points

The disease free, and endemic equilibrium points have been computed in [17] as $\mathcal{E}_0 = (\frac{b}{\mu}, 0, 0, 0, 0)$ and $\mathcal{E}^* = (P^*, M^*, H^*, C^*, R^*)$, where

$$\begin{aligned} P^* &= \frac{b(\mu + \theta + \alpha_3)(\alpha_1 + \alpha_2\mu + \delta_1)(\gamma_1 + \mu + \delta_2)(\mu + \gamma_2)}{\tilde{\Lambda}}, \\ M^* &= \frac{b(\mu + \theta + \alpha_3)(\alpha_1 + \alpha_2\mu + \delta_1)(\gamma_1 + \mu + \delta_2)\Delta^*}{\tilde{\Lambda}}, \\ H^* &= \frac{b(\gamma_1 + \mu + \delta_2)(\mu + \gamma_2)\Delta^*}{\tilde{\Lambda}}, \\ C^* &= \frac{b\theta\alpha_1(\mu + \gamma_2)\Delta^*}{\tilde{\Lambda}}, \\ R^* &= \frac{b(\tilde{\Lambda}_0 - (\mu + \theta + \alpha_3)(\alpha_1 + \alpha_2\mu + \delta_1)(\gamma_1 + \mu + \delta_2)(\mu + \gamma_2))\Delta^*}{\gamma_2\tilde{\Lambda}}, \end{aligned}$$

while

$$\begin{aligned} \Delta^* &= \frac{\mu(\mu + \theta + \alpha_3)(\alpha_1 + \alpha_2\mu + \delta_1)(\gamma_1 + \mu + \delta_2)(\mu + \gamma_2)(\mathcal{R}_0 - 1)}{\tilde{\Lambda}_0 - \theta(\mu + \gamma_2)(\alpha_1\delta_2 + \delta_1(\gamma_1 + \mu + \delta_2))}, \tilde{\Lambda} \\ &= \tilde{\Lambda}_0\Delta^* + \mu(\mu + \theta + \alpha_3)(\alpha_1 + \alpha_2\mu + \delta_1)(\gamma_1 + \mu + \delta_2)(\mu + \gamma_2)\tilde{\Lambda}_0 \\ &= (\theta\alpha_1\gamma_1 + \theta\alpha_2(\gamma_1 + \mu + \delta_2) + \alpha_3(\alpha_1 + \alpha_2\mu + \delta_1)(\gamma_1 + \mu + \delta_2))\gamma_2 + (\mu + \theta + \alpha_3)(\alpha_1 + \alpha_2\mu + \delta_1)(\gamma_1 + \mu + \delta_2)(\mu + \gamma_2). \end{aligned}$$

3.3. The Basic Reproduction Number ($\mathcal{R}_0$)

The $\mathcal{R}_0$ describe the average increase of a phenomenon due to individual in the total population. Here, $\mathcal{R}_0$ represent basic alcohol-abuse initiation number. In [17], the next-generation-matrix method is applied to find $\mathcal{R}_0$. The value of $\mathcal{R}_0$ is given by the following relation. (see Table 1)

$$\mathcal{R}_0 = \frac{\beta_1}{\mu + \theta + \alpha_3} + \frac{\beta_1\theta}{(\mu + \theta + \alpha_3)(\alpha_1 + \alpha_2\mu + \delta_1)}.$$

It has been also shown that the consider model at alcohol-abuse-free equilibrium is stable if $\mathcal{R}_0 \leq 1$ and unstable if $\mathcal{R}_0 > 1$. Further, the alcohol abuse-persistent equilibrium locally asymptotically stable $\mathcal{R}_0 > 1$ and unstable $\mathcal{R}_0 < 1$, for detail see [16]. Here, we present the 3D profile of $\mathcal{R}_0$ in Fig. 1. We use the numerical values given in Table 2.

Table 1
Detail of various parameters.

b	Rate of recruitment into the population is into potential drinkers
$\lambda = \frac{\beta_1 M(\eta) + \beta_2 H(\eta)}{N(\eta)}$	Rate of force of initiation of potential drinker
β_1	Probability that $P(\eta)$ start drinking due to interaction with $M(\eta)$
β_2	Probability that $P(\eta)$ start drinking due to interaction with $H(\eta)$
θ	Rate at which compartment $M(\eta)$ becomes $H(\eta)$
α_1	Rate which heavy drinkers develop complications
α_2	Rate at which heavy drinkers recovers
α_3	Rate at which compartment $M(\eta)$ recovers
γ_1	Rate of recovery of heavy drinkers with induced complications
γ_2	Rate of recovered individuals rejoin the potential drinkers
μ	Represent natural death rate
δ_1	Alcohol-induced death rate In $H(\eta)$
δ_2	rate of death due to complications from alcohol-abuse

3.4. Existence Theory

In this section, we establish some sufficient conditions for the existence and uniqueness of solution of our proposed model. For this, we put the right sides of (4) as

$$\begin{cases} \phi_1(\eta, P, M, H, C, R) = b + \gamma_2 R - (\lambda + \mu)P, \\ \phi_2(\eta, P, M, H, C, R) = \lambda P - (\mu + \theta + \alpha_3)M, \\ \phi_3(\eta, P, M, H, C, R) = \theta M - (\alpha_1 + \alpha_2 + \mu + \delta_1)H, \\ \phi_4(\eta, P, M, H, C, R) = \alpha_1 H - (\gamma_1 + \mu + \delta_2)C, \\ \phi_5(\eta, P, M, H, C, R) = \alpha_2 H + \alpha_3 M + \gamma_1 C - (\mu + \gamma_2)R. \end{cases} \quad (8)$$

Utilizing (8), model (4), with $r \in (0, 1]$ may be express as:

$$\begin{cases} \mathcal{F}_{\eta_0}^r \mathcal{Y}(\eta) = \mathcal{X}(\eta, \mathcal{Y}(\eta)), \quad \eta \in [\eta_0, \mathbf{T}], \\ \mathcal{Y}(0) = \mathcal{Y}_0, \end{cases} \quad (9)$$

where

Table 2
Nomenclatures and numerical values.

Nomenclature	Numerical values
b	65 [16]
β_1	0.020 [16]
β_2	0.010 [assumed]
θ	0.14 [assumed]
α_1	0.02 [16]
α_2	0.001 [16]
α_3	0.0001 [assumed]
γ_1	0.0001 [16]
γ_2	0.0002 [assumed]
μ	0.065 [16]
δ_1	0.002 [16]
δ_2	0.002 [16]

$$\mathcal{Y}(\eta) = \begin{cases} P(\eta), \\ M(\eta), \\ H(\eta), \\ C(\eta), \\ R(\eta). \end{cases}, \quad \mathcal{Y}_0(\eta) = \begin{cases} P_0, \\ M_0, \\ H_0, \\ C_0, \\ R_0, \end{cases} \quad \mathcal{X}(\eta, \mathcal{Y}(\eta)) = \begin{cases} \phi_1(\eta, P, M, H, C, R), \\ \phi_2(\eta, P, M, H, C, R), \\ \phi_3(\eta, P, M, H, C, R), \\ \phi_4(\eta, P, M, H, C, R), \\ \phi_5(\eta, P, M, H, C, R). \end{cases} \quad (10)$$

On applying $\mathbf{I}_0^r$ to (9) yields

$$\mathcal{Y}(\eta) = \mathcal{Y}_0 + \int_0^\eta (\eta - \eta_0)^{r-1} \mathcal{X}(\eta, \mathcal{Y}(\eta)) d\eta. \quad (11)$$

Let $\mathbf{B} = \mathbf{C}[\eta_0, \mathbf{T}]$ be Banach space under the norm $\|\mathcal{Y}\| = \max_{\eta \in [\eta_0, \mathbf{T}]} |\mathcal{Y}(\eta)|$. Additionally, $\mathbb{Q} : \mathbf{B} \rightarrow \mathbf{B}$ be the operator defined by

$$\mathbb{Q}(\mathcal{Y})(\eta) = \mathcal{Y}_0 + \int_0^\eta (\eta - \eta_0)^{r-1} \mathcal{X}(\eta, \mathcal{Y}(\eta)) d\eta. \quad (12)$$

Let the following assumptions hold.

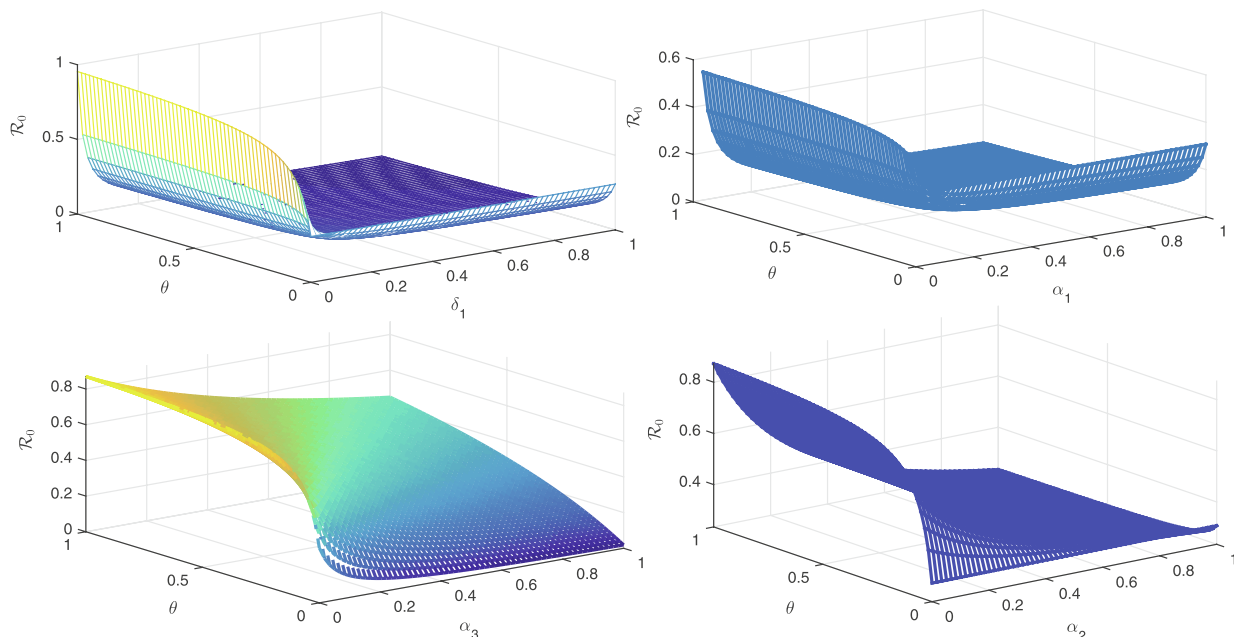


Fig. 1. 3D profile of $\mathcal{R}_0$ of the proposed model.

(H₁)If there exists constants $C_{\mathcal{X}}, M_{\mathcal{X}} > 0$, such that

$$|\mathcal{X}(\eta, \mathcal{Y}(\eta))| \leq C_{\mathcal{X}} |\mathcal{Y}(\eta)| + M_{\mathcal{X}}.$$

(H₂)There exist constants $A_{\mathcal{X}}$, such that for $\mathcal{Y}, \overline{\mathcal{Y}} \in B$ with

$$|\mathcal{X}(\eta, \mathcal{Y}(\eta)) - \mathcal{X}(\eta, \overline{\mathcal{Y}}(\eta))| \leq A_{\mathcal{X}} |\mathcal{Y} - \overline{\mathcal{Y}}|.$$

Theorem 3.2. By assumption (H₁), the problem (9) has at least one solution.

Proof. Let $D = \{\mathcal{Y} \in B : \|\mathcal{Y}\| \leq \rho\}$ be closed and bounded subset of B with

$$r \geq \frac{T^r [C_{\mathcal{X}} \rho + M_{\mathcal{X}}]}{r}.$$

Now, for all $\mathcal{Y} \in D$, one has

$$\begin{aligned} \|\mathcal{Q}(\mathcal{Y})\| &= \max_{\eta \in [\eta_0, T]} \left| \mathcal{Y}_0 + \int_{\eta_0}^{\eta} (\psi - \eta_0)^{r-1} \mathcal{X}(\psi, \mathcal{Y}(\psi)) d\psi \right| \\ &\leq |\mathcal{Y}_0| + \int_{\eta_0}^{\eta} (\psi - \eta_0)^{r-1} |\mathcal{X}(\psi, \mathcal{Y}(\psi))| d\psi \\ &\leq \frac{T^r [C_{\mathcal{X}} \rho + M_{\mathcal{X}}]}{r} \\ &\leq \rho. \end{aligned} \quad (13)$$

Hence, from (13) we have $\|\mathcal{Q}(\mathcal{Y})\| \leq \rho$. So $\mathcal{Q}(D) \subset D$. Continuity of $\mathcal{X}$ means that $\mathcal{Q}$ is continuous. In addition, let $\eta_1 < \eta_2 \in [\eta_0, T]$, such that

$$\begin{aligned} |\mathcal{Q}(\mathcal{Y})(\eta_2) - \mathcal{Q}(\mathcal{Y})(\eta_1)| &= \left| \int_{\eta_0}^{\eta_2} (\eta_2 - \psi)^{r-1} \mathcal{X}(\psi, \mathcal{Y}(\psi)) d\psi \right. \\ &\quad \left. - \int_{\eta_0}^{\eta_1} (\eta_1 - \psi)^{r-1} \mathcal{X}(\psi, \mathcal{Y}(\psi)) d\psi \right| \\ &\leq \frac{C_{\mathcal{X}} \rho + M_{\mathcal{X}}}{r} [(\eta_2 - \eta_0)^r - (\eta_1 - \eta_0)^r]. \end{aligned} \quad (14)$$

The right side of (14) tends to zero with $\eta_2 \rightarrow \eta_1$. So

$$|\mathcal{Q}(\mathcal{Y})(\eta_2) - \mathcal{Q}(\mathcal{Y})(\eta_1)| \rightarrow 0 \text{ as } \eta_2 \rightarrow \eta_1.$$

Thus $\mathcal{Q}$ is equi-continuous. Since $\mathcal{Q}$ is also bounded, hence

$$\|\mathcal{Q}(\mathcal{Y})(\eta_2) - \mathcal{Q}(\mathcal{Y})(\eta_1)\| \rightarrow 0 \text{ as } \eta_2 \rightarrow \eta_1$$

guaranty uniform continuity. Therefore, the consider operator is relatively compact so it is completely continuous. Hence Schauder's fixed point theorem guaranty that the operator $\mathcal{Q}$ has one fixed point at least, it means that the consider problem (9), has one solution at least.

Theorem 3.3. Let the assumption (H₂) and the condition $\Theta = \frac{A_{\mathcal{X}} T^r}{r} < 1$ is hold, then problem (9) has unique solution.

Proof. Consider $\mathcal{Y}, \overline{\mathcal{Y}} \in B$, then we have

$$\|\mathcal{Q}(\mathcal{Y}) - \mathcal{Q}(\overline{\mathcal{Y}})\| \leq \frac{A_{\mathcal{X}} T^r}{r} \|\mathcal{Y} - \overline{\mathcal{Y}}\| \quad (15)$$

From (15), one has

$$\|\mathcal{Q}(\mathcal{Y}) - \mathcal{Q}(\overline{\mathcal{Y}})\| \leq \Theta \|\mathcal{Y} - \overline{\mathcal{Y}}\|.$$

Hence, $\mathcal{Q}$ is a contraction. Therefore, model (9) has unique solution (Banach's contraction principle).

4. Ulam-Hyer's Stability

In this section, we will establish sufficient conditions for UH type stability. This type of stability is important for the numerical solution of a problem. The basic question about the approximate solution is "under what conditions the approximate solution of a problem is nearly equal to the exact solution". This important question has been successfully answered by Ulam in 1940 [45]. The mentioned stability was further studied and generalized by Hyer's and Rassias to UH, UH-Rassias stability [46,47]. For definition and some discussion on the stated stability one may read fundamental concepts in these references [41–44,48].

Consider the corresponding perturb problem of system (9) as

$$\begin{cases} \mathcal{T}_{\eta_0}^r \mathcal{Y}(\eta) = \mathcal{X}(\eta, \mathcal{Y}(\eta)) + \varphi(\eta), \quad \eta \in [\eta_0, T], \\ \mathcal{Y}(0) = \mathcal{Y}_0, \end{cases} \quad (16)$$

Here, $\varphi(\eta) \in C([\eta_0, T], R)$ with property $|\varphi(\eta)| \leq \epsilon$, for $\epsilon > 0$. Also for any solution, $\mathcal{Y} \in B$ of perturb problem (16) such that,

$$\mathcal{Q}\mathcal{Y}(\eta) = \mathcal{Y}(\eta) + \varphi(\eta).$$

From the solution of above perturb problem, we can write

$$\|\mathcal{Q}\mathcal{Y}(\eta) - \mathcal{Y}(\eta)\| \leq \nabla \epsilon, \text{ where } \nabla = \frac{T^r}{r}. \quad (17)$$

Theorem 4.1. The solution of system (9) is UH and generalized UH stable if $A_{\mathcal{X}} \nabla < 1$.

Proof. Let $\overline{\mathcal{Y}}(\eta) \in B$ be any solution of problem (16) and $\mathcal{Y} \in X$ represent unique solution to problem (9), then

$$\begin{aligned} |\mathcal{Y}(\eta) - \overline{\mathcal{Y}}(\eta)| &= |\mathcal{Y}(\eta) - \mathcal{Q}\overline{\mathcal{Y}}(\eta)| \leq |\mathcal{Y}(\eta) - \mathcal{Q}\mathcal{Y}(\eta)| + |\mathcal{Q}\mathcal{Y}(\eta) - \mathcal{Q}\overline{\mathcal{Y}}(\eta)| \\ &\leq \nabla \epsilon + A_{\mathcal{X}} \nabla |\mathcal{Y}(\eta) - \overline{\mathcal{Y}}(\eta)| \\ &\leq \frac{\nabla \epsilon}{1 - A_{\mathcal{X}} \nabla}. \end{aligned} \quad (18)$$

Hence, problem (9) is UH and generalized UH type stable. In this way we can conclude that the solution of the considered model (4) is UH and generalized UH stable. On the same way the remaining types of the stated stability can be studied.

5. Numerical Scheme for the Proposed Model (4)

In this section, we are going to approximate the solution to our consider problem under CFOD. For the desired solution we will use Euler's method which is the simplest method for the solution of different problems. Euler's method for an initial value problem under conformable derivative has already been studied in some papers like [51]. Here, we follow the same procedure to extend the proposed model as

$$\begin{cases} \mathcal{T}_{\eta}^r \mathcal{U}(\eta) = \mathbf{H}(\eta, \mathcal{U}), \\ \mathcal{U}(0) = \mathcal{U}_0, \end{cases}$$

on interval of study $[a, b]$, with $h = \frac{b-a}{n}$, is given by

$$\mathcal{U}_{n+1} = \mathcal{U}_0 + \frac{h^r}{r} \mathbf{I}_{\eta}^r \mathbf{H}(\eta_n, \mathcal{U}_n).$$

Using the above definition, the approximate solution of model (4) is calculated as,

$$\begin{cases} P_{n+1} = P_n + \frac{h^r}{r} \mathbf{I}_\eta^r \left(b + \gamma_2 R_n - \left(\frac{\beta_1 M_n + \beta_2 H_n}{N_n} \right) P_n - \mu P_n \right), \\ M_{n+1} = M_n + \frac{h^r}{r} \mathbf{I}_\eta^r \left(\left(\frac{\beta_1 M_n + \beta_2 H_n}{N_n} \right) P_n - (\mu + \theta + \alpha_3) M_n \right), \\ H_{n+1} = H_n + \frac{h^r}{r} \mathbf{I}_\eta^r (\theta M_n - (\alpha_1 + \alpha_2 + \mu + \delta_1) H_n), \\ C_{n+1} = C_n + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_1 H_n - (\gamma_1 + \mu + \delta_2) C_n), \\ R_{n+1} = R_n + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_2 H_n + \alpha_3 M_n + \gamma_1 C_n - (\mu + \gamma_2) R_n), \end{cases} \quad (19)$$

where $n = 0, 1, 2, \dots$, with initial conditions as

$$P(0) = P_0, \quad M(0) = M_0, \quad H(0) = H_0, \quad C(0) = C_0, \quad R(0) = R_0. \quad (20)$$

Now let's calculate some terms of the solution. Put $n = 0$ in (19) and using initial conditions (20), we get the first term as:

$$\begin{cases} P_1 = P_0 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(b + \gamma_2 R_0 - \left(\frac{\beta_1 M_0 + \beta_2 H_0}{N_0} \right) P_0 - \mu P_0 \right), \\ M_1 = M_0 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(\left(\frac{\beta_1 M_0 + \beta_2 H_0}{N_0} \right) P_0 - (\mu + \theta + \alpha_3) M_0 \right), \\ H_1 = H_0 + \frac{h^r}{r} \mathbf{I}_\eta^r (\theta M_0 - (\alpha_1 + \alpha_2 + \mu + \delta_1) H_0), \\ C_1 = C_0 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_1 H_0 - (\gamma_1 + \mu + \delta_2) C_0), \\ R_1 = R_0 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_2 H_0 + \alpha_3 M_0 + \gamma_1 C_0 - (\mu + \gamma_2) R_0), \end{cases} \quad (21)$$

For second term put $n = 1$ in (19), we have

$$\begin{cases} P_2 = P_1 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(b + \gamma_2 R_1 - \left(\frac{\beta_1 M_1 + \beta_2 H_1}{N_1} \right) P_1 - \mu P_1 \right), \\ M_2 = M_1 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(\left(\frac{\beta_1 M_1 + \beta_2 H_1}{N_1} \right) P_1 - (\mu + \theta + \alpha_3) M_1 \right), \\ H_2 = H_1 + \frac{h^r}{r} \mathbf{I}_\eta^r (\theta M_1 - (\alpha_1 + \alpha_2 + \mu + \delta_1) H_1), \\ C_2 = C_1 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_1 H_1 - (\gamma_1 + \mu + \delta_2) C_1), \\ R_2 = R_1 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_2 H_1 + \alpha_3 M_1 + \gamma_1 C_1 - (\mu + \gamma_2) R_1), \end{cases} \quad (22)$$

On the same way we can calculate the other terms for $n = 2, 3$ as,

$$\begin{cases} P_3 = P_2 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(b + \gamma_2 R_2 - \left(\frac{\beta_1 M_2 + \beta_2 H_2}{N_2} \right) P_2 - \mu P_2 \right), \\ M_3 = M_2 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(\left(\frac{\beta_1 M_2 + \beta_2 H_2}{N_2} \right) P_2 - (\mu + \theta + \alpha_3) M_2 \right), \\ H_3 = H_2 + \frac{h^r}{r} \mathbf{I}_\eta^r (\theta M_2 - (\alpha_1 + \alpha_2 + \mu + \delta_1) H_2), \\ C_3 = C_2 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_1 H_2 - (\gamma_1 + \mu + \delta_2) C_2), \\ R_3 = R_2 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_2 H_2 + \alpha_3 M_2 + \gamma_1 C_2 - (\mu + \gamma_2) R_2), \end{cases} \quad (23)$$

and

$$\begin{cases} P_4 = P_3 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(b + \gamma_2 R_3 - \left(\frac{\beta_1 M_3 + \beta_2 H_3}{N_3} \right) P_3 - \mu P_3 \right), \\ M_4 = M_3 + \frac{h^r}{r} \mathbf{I}_\eta^r \left(\left(\frac{\beta_1 M_3 + \beta_2 H_3}{N_3} \right) P_3 - (\mu + \theta + \alpha_3) M_3 \right), \\ H_4 = H_3 + \frac{h^r}{r} \mathbf{I}_\eta^r (\theta M_3 - (\alpha_1 + \alpha_2 + \mu + \delta_1) H_3), \\ C_4 = C_3 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_1 H_3 - (\gamma_1 + \mu + \delta_2) C_3), \\ R_4 = R_3 + \frac{h^r}{r} \mathbf{I}_\eta^r (\alpha_2 H_3 + \alpha_3 M_3 + \gamma_1 C_3 - (\mu + \gamma_2) R_3), \end{cases} \quad (24)$$

and soon. The accuracy can be improved by choosing h very small.

6. Numerical presentation and discussion

Here, we present the considered model solution graphically via using some real data given as

On using the numerical values given in Table 2 to simulate the results for various fractional orders in Figs. 2–6. We use two sets of fractional orders values to investigate the effect of fractional orders on the dynamics of the phenomenon. We see from Fig. 2 that the population density of potential drinker of alcohol is increasing with different scenario because of variation in fractional order. In the same way, we see from Fig. 3 that the population density of moderate alcohol abusers is decaying with the passage of time. In addition, in Fig. 4 one can observe the rapid growth in heavy alcohol abusers and then decline in the transmission dynamics of the said class. Also, Fig. 5 and 6 show the slowly decay in population density of heavy alcohol abusers with induced complications, and recovered class population with different fractional order. Moreover, the UH stability can be observed in the dynamics of all classes. Now in Figs. 7–11, we present the approximate

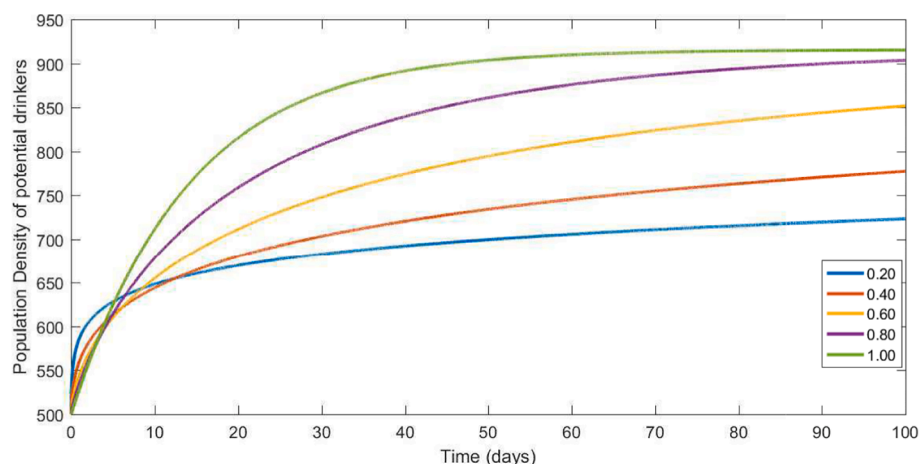


Fig. 2. Approximate solution presentation of population density of potential drinkers graphically using different fractional orders.

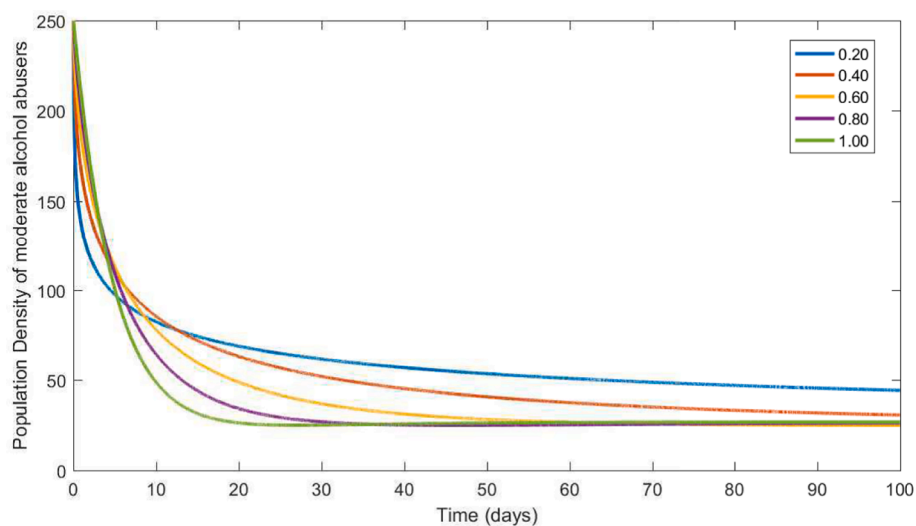


Fig. 3. Approximate solution presentation of population density of moderate alcohol abusers graphically using different fractional orders.

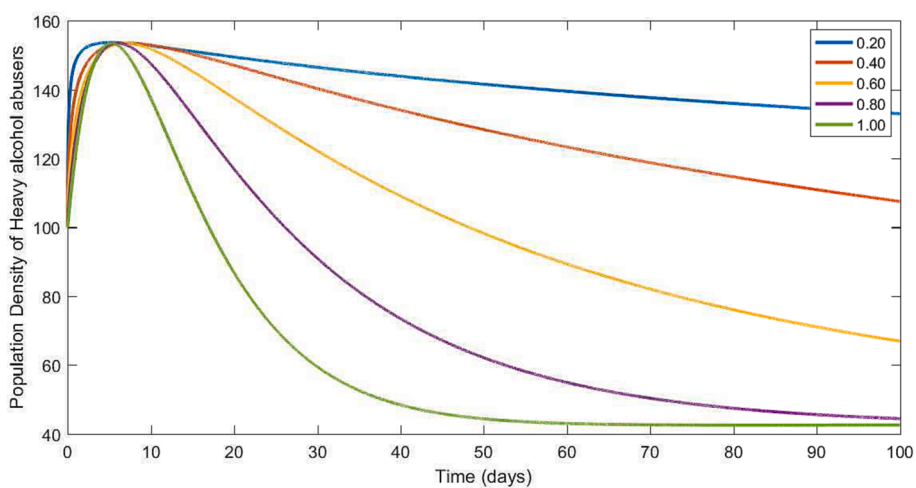


Fig. 4. Approximate solution presentation of population density of heavy alcohol abusers graphically using different fractional orders.

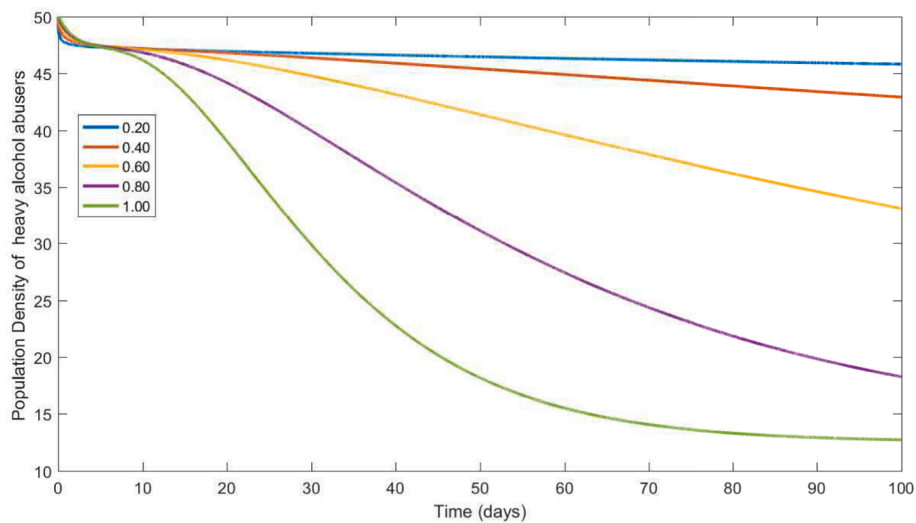


Fig. 5. Approximate solution presentation of population density of heavy alcohol abusers with induced complications graphically using different fractional orders.

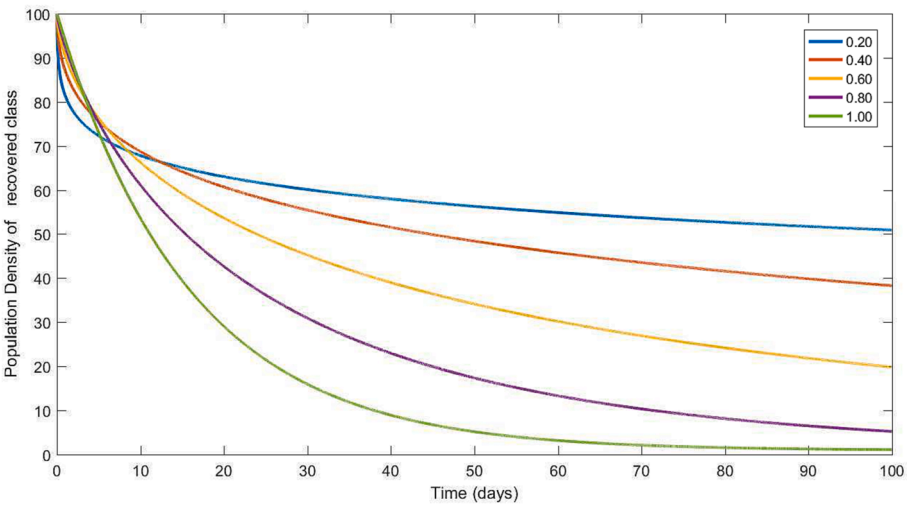


Fig. 6. Approximate solution presentation of population density of recovered class graphically using different fractional orders.

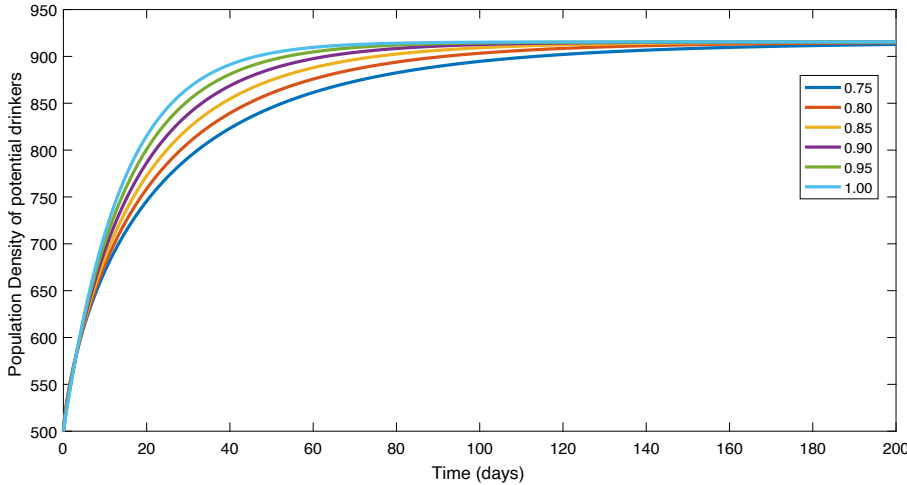


Fig. 7. Approximate solution presentation of population density of potential drinkers graphically using different fractional orders.

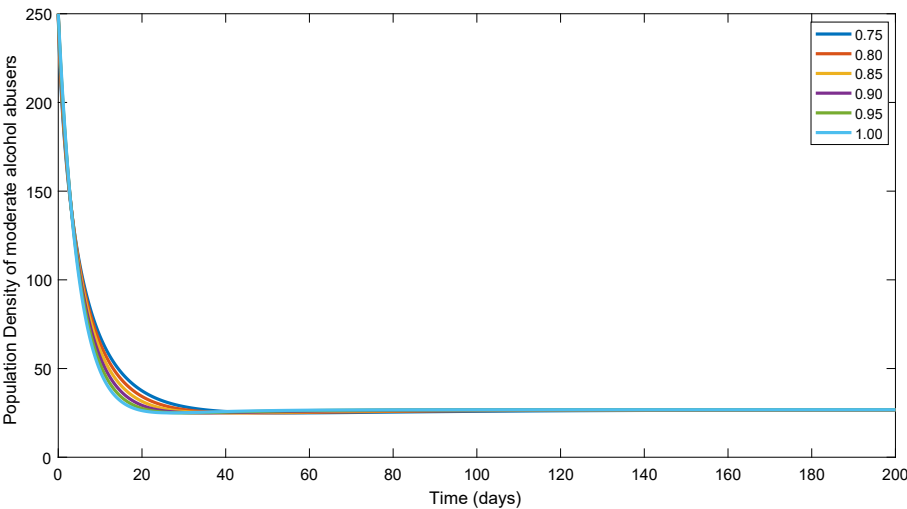


Fig. 8. Approximate solution presentation of population density of moderate alcohol abusers graphically using different fractional orders.

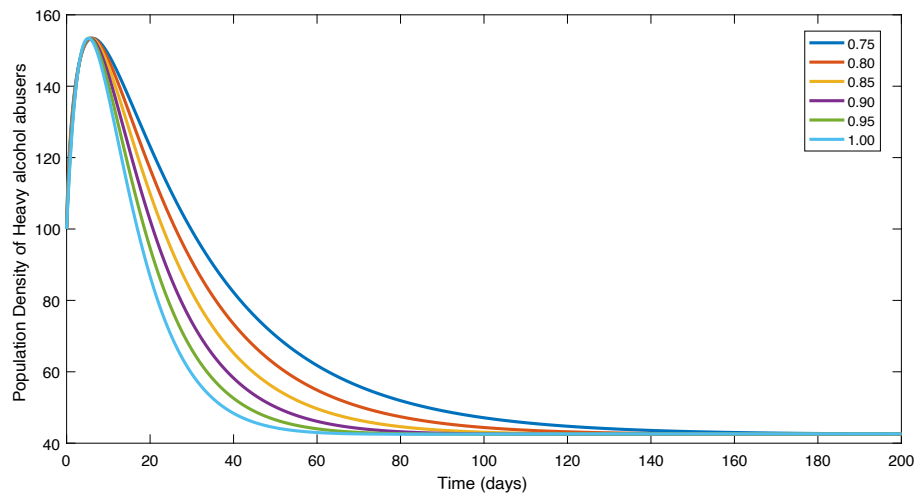


Fig. 9. Approximate solution presentation of population density of Heavy alcohol abusers graphically using different fractional orders.

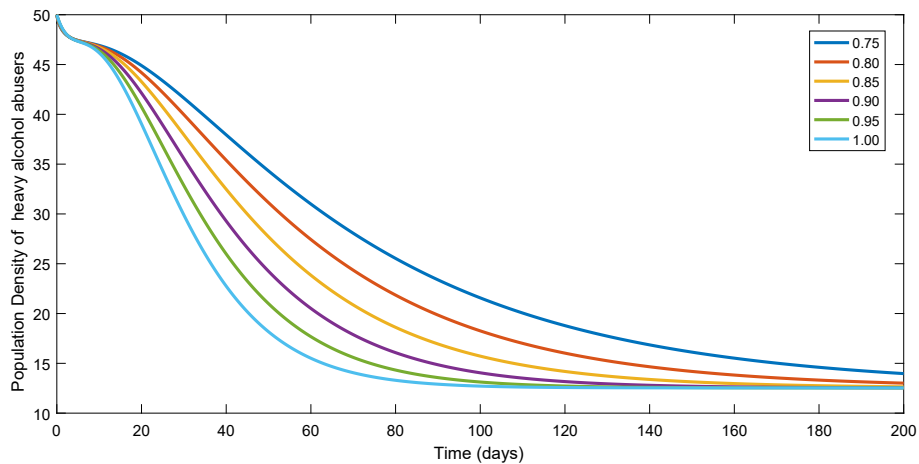


Fig. 10. Approximate solution presentation of population density of heavy alcohol abusers with induced complications graphically using different fractional orders.

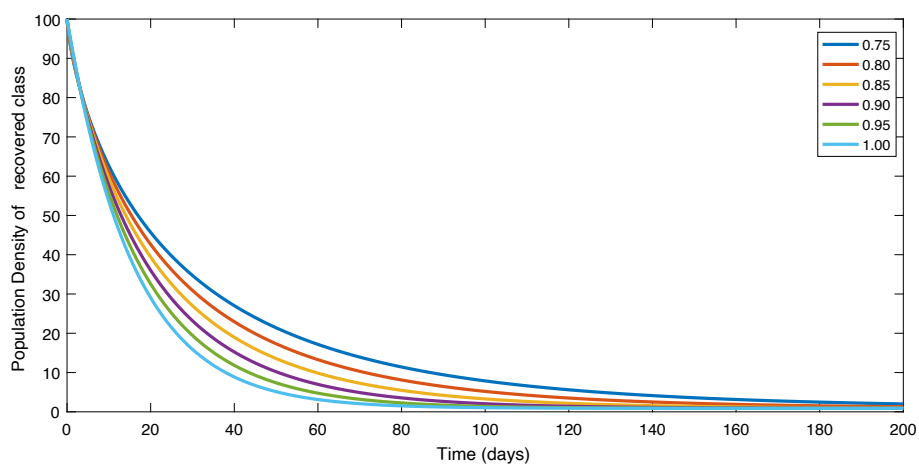


Fig. 11. Approximate solution presentation of population density of recovered class graphically using different fractional orders.

solutions of various compartment of the proposed model graphically for different fractional orders. In addition, we have considered slightly greater fractional values of the order of derivatives. Here, we see that the corresponding convergent can be rapidly achieved by increasing the value of the fractional orders. Here, on greater fractional orders we see

the growth in the potential drinker population in Fig. 7. Also here, we observe convergence in the solution. Onward in other compartments dynamics, one can observe the same behaviors as discussed earlier with convergent behavior. Hence the solution is Uh stable also and behaves like classical order results which has the interesting feature of CFOD.

Here, we remark that the transmission of acholic abuse in society has been demonstrated by using some new analysis based on conformable fractional order derivatives. We see that the suggested derivative has also powerful effect in the description of real world problems to understand the transmission dynamics in society.

7. Concluding remarks

This research work is devoted to establish qualitative and numerical analysis to five compartments alcohol-abuse model. The proposed model has investigated in the the sense of conformable fractional order derivative. We have proved sufficient condition for the existence of at least one approximate solution using fixed point theory. In addition, stability result has also established for the corresponding model in the sense of Ulam-Hyers. Here, for numerical purpose, we have used Euler method. Using some real values of the parameters, we have presented the approximate numerical solution graphically against different fractional order. From graphical presentation, we see that conformable derivative has also a powerful tool to use to describe various real world problems. In future, we will apply the aforementioned analysis in other complicated dynamical systems under the CFOD for more detailed investigations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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